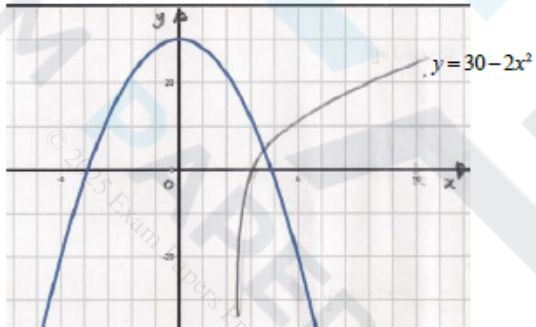


Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$f(3.5) = -4.8, f(4) = (+)3.1$	M1	1.1b
	Change of sign and function continuous in interval $[3.5, 4] \Rightarrow \text{Root}^*$	A1*	2.4
		(2)	
(b)	Attempts $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow x_1 = 4 - \frac{3.099}{16.67}$	M1	1.1b
	$x_1 = 3.81$	A1	1.1b
	$y = \ln(2x - 5)$	(2)	
(c)		M1	3.1a
	Attempts to sketch both $y = \ln(2x - 5)$ and $y = 30 - 2x^2$		
	States that $y = \ln(2x - 5)$ meets $y = 30 - 2x^2$ in just one place, therefore $y = \ln(2x - 5) = 30 - 2x^2$ has just one root $\Rightarrow f(x) = 0$ has just one root	A1	2.4
		(2)	
(6 marks)			

**Notes:**

(a)	
M1:	Attempts $f(x)$ at both $x = 3.5$ and $x = 4$ with at least one correct to 1 significant figure
A1*:	$f(3.5)$ and $f(4)$ correct to 1 sig figure (rounded or truncated) with a correct reason and conclusion. A reason could be change of sign, or $f(3.5) \times f(4) < 0$ or similar with $f(x)$ being continuous in this interval. A conclusion could be 'Hence root' or 'Therefore root in interval'
(b)	
M1:	Attempts $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ evidenced by $x_1 = 4 - \frac{3.099}{16.67}$
A1:	Correct answer only $x_1 = 3.81$
(c)	
M1:	For a valid attempt at showing that there is only one root. This can be achieved by • Sketching graphs of $y = \ln(2x - 5)$ and $y = 30 - 2x^2$ on the same axes • Showing that $f(x) = \ln(2x - 5) + 2x^2 - 30$ has no turning points • Sketching a graph of $f(x) = \ln(2x - 5) + 2x^2 - 30$
A1:	Scored for correct conclusion

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = x^x \Rightarrow \ln y = x \ln x$	M1	This mark is for a method to find the x -coordinate of the turning point of C by taking logarithms
	$\ln y = x \ln x \Rightarrow \frac{1}{y} \frac{dy}{dx} = \ln x + 1$	M1	This mark is given for a method using implicit differentiation
		A1	This mark is given for a correct expression for $\frac{1}{y} \frac{dy}{dx}$
	Setting $\frac{dy}{dx} = 0$, $\ln x + 1 = 0$	M1	This mark is given for a method for finding the turning point of C by setting $\frac{dy}{dx} = 0$
	$x = e^{-1}$	A1	This mark is given for correctly finding a value for the x -coordinate of the turning point of C

(b)	$1.5^{1.5} = 1.837\dots, 1.6^{1.6} = 2.121\dots$	M1	This mark is given for substituting 1.5 and 1.6 into $y = x^x$
	The curve C contains the points (1.5, 1.8) and (1.6, 2.1). At $P, y = 2$ Since C is continuous, $1.5 < \alpha < 1.6$	A1	This mark is given for a valid explanation that C contains the points (1.5, 1.8) and (1.6, 2.1) and is continuous
(c)	$x_1 = 1.5$ $x_2 = 2 \times 1.5^{-0.5} = 1.633$	M1	This mark is given for finding a correct value for x_2
	$x_3 = 2 \times 1.633^{-0.633} = 1.466$ $x_4 = 2 \times 1.466^{-0.466} = 1.673$	A1	This mark is given for finding a correct value for x_4
(d)	For example: x_n oscillates is periodic is non-convergent	B1	This mark is given for a valid statement about the long-term behaviour of x_n
	between 1 and 2	B1	This mark is given for stating that the behaviour is between 1 and 2
(Total 11 marks)			

Q3.

Question	Scheme	Marks	AOs
(a)	$f'(x) = 2x + \frac{4x-4}{2x^2-4x+5}$	M1	1.1b
		A1	1.1b
	$2x + \frac{4x-4}{2x^2-4x+5} = 0 \Rightarrow 2x(2x^2-4x+5) + 4x-4 = 0$	dM1	1.1b
	$2x^3 - 4x^2 + 7x - 2 = 0^*$	A1*	2.1
(b)	(i) $x_2 = \frac{1}{7}(2 + 4(0.3)^2 - 2(0.3)^3)$	M1	1.1b
	$x_2 = 0.3294$	A1	1.1b
	(ii) $x_4 = 0.3398$	A1	1.1b
		(3)	
(c)	$h(x) = 2x^3 - 4x^2 + 7x - 2$ $h(0.3415) = 0.00366\dots \quad h(0.3405) = -0.00130\dots$	M1	3.1a
	States: - there is a change of sign $f'(x)$ is continuous $\alpha = 0.341$ to 3dp	A1	2.4
		(2)	
(9 marks)			
Notes			



(a)

M1: Differentiates $\ln(2x^2 - 4x + 5)$ to obtain $\frac{g(x)}{2x^2 - 4x + 5}$ where $g(x)$ could be 1

A1: For $f'(x) = 2x + \frac{4x - 4}{2x^2 - 4x + 5}$

dM1: Sets their $f'(x) = ax + \frac{g(x)}{2x^2 - 4x + 5} = 0$ and uses "correct" algebra, condoning slips, to obtain a

cubic equation. E.g. Look for $ax(2x^2 - 4x + 5) \pm g(x) = 0$ o.e., condoning slips, followed by some attempt to simplify

A1*: Achieves $2x^3 - 4x^2 + 7x - 2 = 0$ with no errors. (The dM1 mark must have been awarded)

(b)(i)

M1: Attempts to use the iterative formula with $x_1 = 0.3$. If no method is shown award for $x_2 = \text{awrt } 0.33$

A1: $x_2 = \text{awrt } 0.3294$ Note that $\frac{1153}{3500}$ is correct

Condone an incorrect suffix if it is clear that a correct value has been found

(b)(ii)

A1: $x_4 = \text{awrt } 0.3398$ Condone an incorrect suffix if it is clear that a correct value has been found

(c)

M1: Attempts to substitute $x = 0.3415$ and $x = 0.3405$ into a suitable function and gets one value correct (rounded or truncated to 1 sf). It is allowable to use a tighter interval that contains the root 0.340762654

Examples of suitable functions are $2x^3 - 4x^2 + 7x - 2$, $x - \frac{1}{7}(4x^2 - 2x^3 + 2)$ and $f'(x)$ as this has been

found in part (a) with $f'(0.3405) = -0.00067\dots$, $f'(0.3415) = (+) 0.0018$

There must be sufficient evidence for the function, which would be for example, a statement such as $h(x) = 2x^3 - 4x^2 + 7x - 2$ or sight of embedded values that imply the function, not just a value or values

even if both are correct. Condone $h(x)$ being mislabelled as f

$h(0.3415) = 2 \times 0.3415^3 - 4 \times 0.3415^2 + 7 \times 0.3415 - 2$

A1: Requires

- both calculations correct (rounded or truncated to 1sf)
- a statement that there is a change in sign and that the function is continuous
- a minimal conclusion e.g. $\checkmark$, proven, $\alpha = 0.341$, root

Q4.

Question	Scheme	Marks	AOs
(a)	$(f'(x) =) 4 \cos\left(\frac{1}{2}x\right) - 3$	M1 A1	1.1b 1.1b
	Sets $f'(x) = 4 \cos\left(\frac{1}{2}x\right) - 3 = 0 \Rightarrow x =$	dM1	3.1a
	$x = 14.0$ Cao	A1	3.2a
		(4)	
(b)	Explains that $f(4) > 0$, $f(5) < 0$ and the function is continuous	B1	2.4
		(1)	
(c)	Attempts $x_1 = 5 - \frac{8 \sin 2.5 - 15 + 9}{4 \cos 2.5 - 3}$ (NB $f(5) = -1.212...$ and $f'(5) = -6.204...$)	M1	1.1b
	$x_1 = \text{awrt } 4.80$	A1	1.1b
		(2)	
(7 marks)			
Notes:			

(a)

M1: Differentiates to obtain $k \cos\left(\frac{1}{2}x\right) \pm a$ where a is a constant which may be zero and no other terms. The brackets are not required.

A1: Correct derivative $f'(x) = 4 \cos\left(\frac{1}{2}x\right) - 3$. Allow unsimplified e.g. $f'(x) = \frac{1}{2} \times 8 \cos\left(\frac{1}{2}x\right) - 3x^0$

There is no need for $f'(x) = \dots$ or $\frac{dy}{dx} = \dots$ just look for the expression and the brackets are not required.

dM1: For the complete strategy of proceeding to a value for x .

Look for

- $f'(x) = a \cos\left(\frac{1}{2}x\right) + b = 0$, $a, b \neq 0$
- Correct method of finding a valid solution to $a \cos\left(\frac{1}{2}x\right) + b = 0$

Allow for $a \cos\left(\frac{1}{2}x\right) + b = 0 \Rightarrow \cos\left(\frac{1}{2}x\right) = \pm k \Rightarrow x = 2 \cos^{-1}(\pm k)$ where $|k| < 1$

If this working is not shown then you may need to check their value(s).

For example $4 \cos\left(\frac{1}{2}x\right) - 3 = 0 \Rightarrow x = 1.4... \text{ or } 11.1... \text{ (or } 82.8... \text{ or } 637... \text{ or } 803 \text{ in degrees)}$ would indicate this method.

A1: Selects the correct turning point $x = 14.0$ and not just 14 or unrounded e.g. 14.011...

Must be this value only and no other values unless they are clearly rejected or 14.0 clearly selected. Ignore any attempts to find the y coordinate.

Correct answer with no working scores no marks.

(b)

B1: See scheme. Must be a full reason, (e.g. change of sign and continuous)

Accept equivalent statements for $f(4) > 0$, $f(5) < 0$ e.g. $f(4) \times f(5) < 0$, "there is a change of sign", "one negative one positive". A minimum is "change of sign and continuous" but do **not** allow this mark if the comment about continuity is clearly incorrect e.g. "because x is continuous" or "because the interval is continuous".

(c)

M1: Attempts $x_1 = 5 - \frac{f(5)}{f'(5)}$ to obtain a value following through on their $f'(x)$ as long as it is a “changed” function.

Must be a correct N-R formula used – may need to check their values.

Allow if attempted in degrees. For reference in degrees $f(5) = -5.65\dots$ and $f'(5) = 0.996\dots$ and gives $x_1 = 10.67\dots$

There must be clear evidence that $5 - \frac{f(5)}{f'(5)}$ is being attempted.

$$\text{so e.g. } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \Rightarrow x_1 = 4.80 \text{ scores M0 as does e.g. } x_1 = x - \frac{8\sin\left(\frac{1}{2}x\right) - 3x + 9}{4\cos\left(\frac{1}{2}x\right) - 3} = 4.80$$

BUT evidence may be provided by the accuracy of their answer. Note that the full N-R accuracy is 4.804624337 so e.g. 4.805 or 4.804 (truncated) with no evidence of incorrect work may imply the method.

A1: $x_1 = \text{awrt } 4.80$ not awrt 4.8 but isw if awrt 4.80 is seen. Ignore any subsequent iterations.

Note that work for part (a) cannot be recovered in part (c)

Note also:

$$5 - \frac{f(5)}{f'(5)} = \text{awrt } 4.80 \text{ following a correct derivative scores M1A1}$$

$$5 - \frac{f(5)}{f'(5)} \neq \text{awrt } 4.80 \text{ with no evidence that } 5 - \frac{f(5)}{f'(5)} \text{ was attempted scores M0}$$